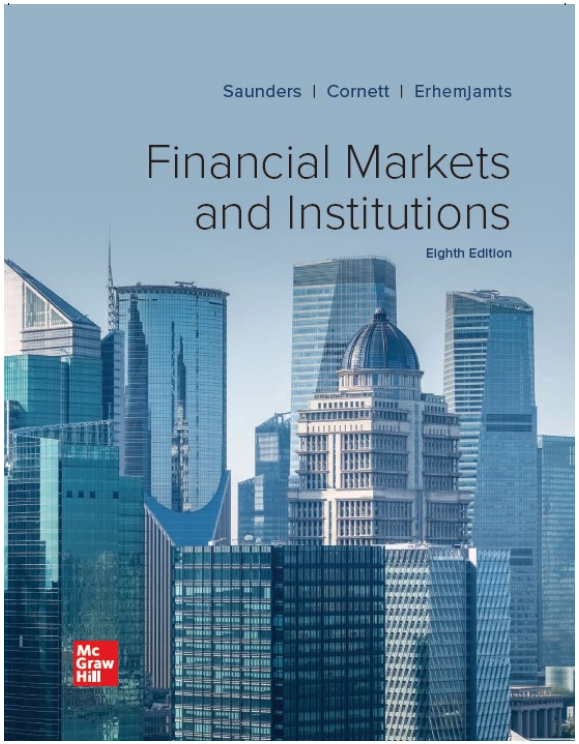
# Chapter Nine

# Foreign Exchange Markets




Copyright © 2022 McGraw-Hill. All rights reserved. No reproduction or distribution without the prior written consent of McGraw-Hill.

# Overview of Foreign Exchange Markets

- Cash flows from the sale of products, services, or assets denominated in a foreign currency are transacted in **foreign exchange (FX) markets**
  - A **foreign exchange rate** is the price at which one currency (e.g., the U.S. dollar) can be exchanged for another currency (e.g., the Swiss franc) in the foreign exchange markets
  - These transactions expose U.S. corporations and investors to **foreign exchange risk** as the cash flows are converted into and out of U.S. dollars
- **Currency depreciation (appreciation)** occurs when a country's currency falls (rises) in value relative to other currencies

© 2022 McGraw-Hill Education.

# Background and History of Foreign Exchange Markets

- During most of the 1800s, FX markets operated under a *gold system*, where currency issuers guaranteed to redeem notes, upon demand, in an equivalent amount of gold
- From 1944-1971, the *Bretton Woods Agreement* called for the exchange rate of one currency for another to be fixed within narrow bands around a specified rate with the help of government intervention
  - *Smithsonian Agreement of 1971* allowed the dollar to be devalued and the boundaries between which exchange rates could fluctuate were increased from 1% to 2.25%
  - In 1973, under *Smithsonian Agreement II*, exchange rate boundaries were eliminated, creating a free-floating system that is still partially in place

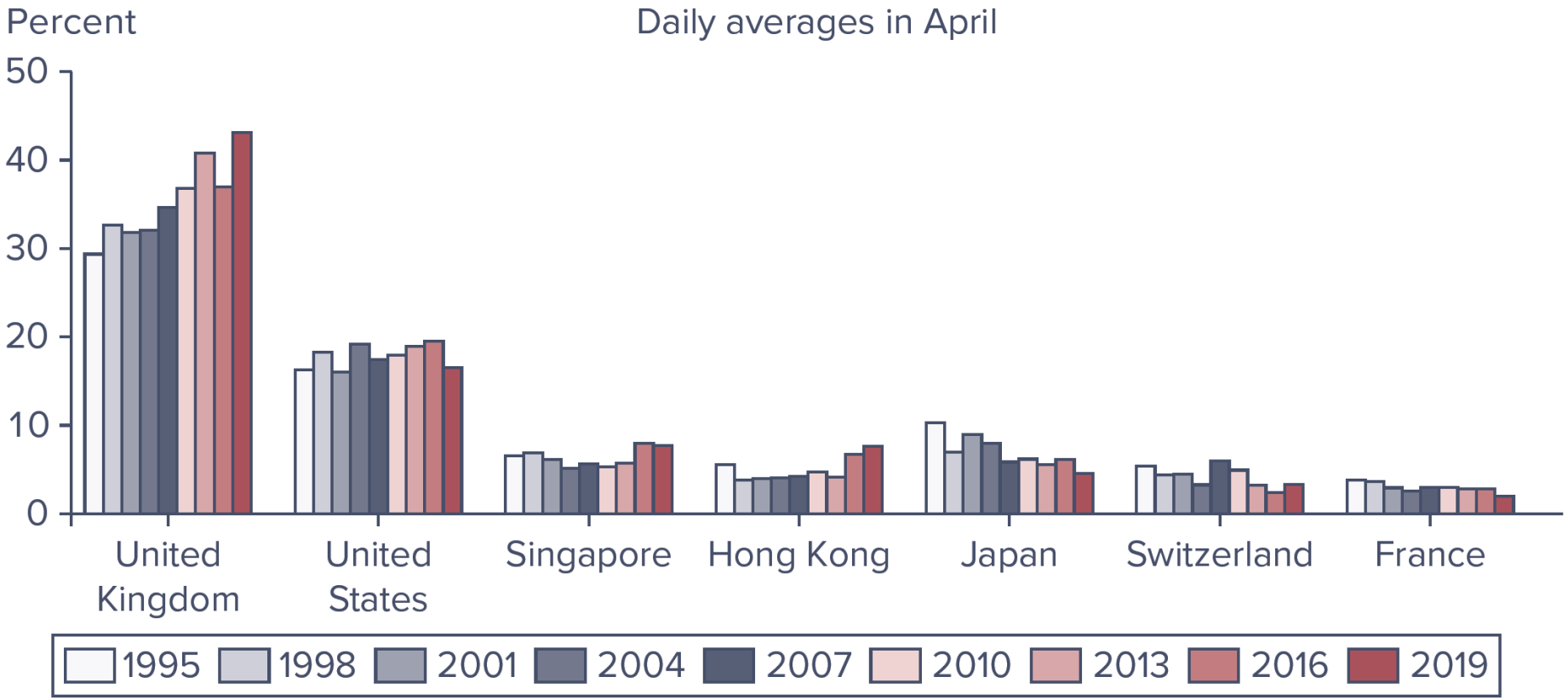
© 2022 McGraw-Hill Education.

# Background and History of Foreign Exchange Markets (Continued)

- Though the free-floating FX rate system is still partially in place, central governments may still intervene in the FX markets
  - In 1992, 12 major European countries and Vatican City pegged their exchange rates together to create a single currency, called the euro
- Until 1972, the interbank FX market was the only channel through which spot and forward FX transactions took place
  - Since 1972, organized markets such as the International Money Market (IMM) or the Chicago Mercantile Exchange (CME) have developed derivatives trading in FX currency futures and options

© 2022 McGraw-Hill Education.

# Largest Global FX Markets




© 2022 McGraw-Hill Education.

# The Introduction of the Euro

- Euro is the European Union's single currency
  - Started trading on January 1, 1999
  - Creation of the euro had its origins in the creation of the European Community (EC) a consolidation of three European communities in 1967:
    - European Coal and Steel Community, the European Economic Market, and the European Atomic Energy Community
  - Significant effect throughout Europe, as well as the global financial system
  - Dollar remains the unparalleled medium of exchange because it is the world's easiest currency to buy or sell
    - In 2019, 88% of all FX transactions were denominated in dollars, while 32% were denominated in euros

© 2022 McGraw-Hill Education.

# Dollarization

- **Dollarization** is the use of a foreign currency in parallel to, or instead of, the local currency
  - After the gold standard and the Bretton Woods Agreement were abandoned, many countries achieved currency stabilization by pegging the local currency to a major convertible currency
  - Other countries abandoned their local currency in favor of exclusive use of the U.S. dollar (or another major international currency, such as the euro)
- Dollarization may occur unofficially or officially
- Major advantage is the promotion of fiscal discipline and thus greater financial stability and lower inflation
  - Panama, Ecuador, and El Salvador are the biggest economies to have officially dollarized

© 2022 McGraw-Hill Education.

# Free-Floating Yuan

- In 2005, China shifted away from its currency's (the yuan) peg to the U.S. dollar
  - China stated the value of the yuan would be determined using a "managed" floating system with reference to an unspecified basket of foreign currencies
  - Partial free-floating of the yuan was in part the result of pressure from Western countries whose politicians argued that China's currency regime gave it an unfair advantage in global markets
- In 2009, China began a pilot program of internationalizing its currency by allowing Hong Kong banks to trade the yuan
  - In 2011, China began to allow Americans to trade in the currency
- In 2015, the IMF designated the Chines yuan an IMF-accepted reserve currency

© 2022 McGraw-Hill Education.

# Foreign Exchange Rates

- Recall, a FX rate is the price at which one currency (e.g., the U.S. dollar) can be exchanged for another currency (e.g., the Swiss franc)

- FX rates are listed in two ways:
  1. U.S. dollars received for one unit of the foreign currency exchanged (IN US$ or USD, also referred to as the *direct quote*)
  2. Foreign currency received for each U.S. dollar exchanged (PER US$, also referred to as the *indirect quote*)

© 2022 McGraw-Hill Education.

# Foreign Currency Exchange Rates

**Exchange Rates: New York Closing Snapshot**
**Friday, March 13, 2020**
**U.S.-dollar foreign-exchange rates in late New York trading**

| Country/Currency | IN US$ | | PER US$ | |
|---|---|---|---|---|
| | Fri | Thurs | Fri | Thurs |
| **Americas** | | | | |
| Argentina peso | 0.0159 | 0.0159 | 62.90 | 62.70 |
| Brazil real | 0.2058 | 0.2086 | 4.8595 | 4.7941 |
| Canada dollar | 0.7243 | 0.7181 | 1.3807 | 1.3926 |
| 1-month forward | 0.7148 | 0.7209 | 1.3990 | 1.3872 |
| 3-month forward | 0.7149 | 0.7210 | 1.3988 | 1.3870 |
| 6-month forward | 0.7151 | 0.7211 | 1.3984 | 1.3868 |
| Chile peso | 0.001192 | 0.001176 | 839.00 | 850.60 |
| Colombia peso | 0.000249 | 0.000248 | 4023.01 | 4028.00 |
| Ecuador US dollar | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| Mexico peso | 0.0456 | 0.0456 | 21.9168 | 21.9202 |
| Uruguay peso | 0.02304 | 0.02314 | 43.41 | 43.21 |
| **Asia-Pacific** | | | | |
| Australian dollar | 0.6193 | 0.6237 | 1.6147 | 1.6033 |
| China yuan | 0.1427 | 0.1423 | 7.0082 | 7.0296 |
| Hong Kong dollar | 0.1287 | 0.1286 | 7.7695 | 7.7773 |
| India rupee | 0.01354 | 0.01341 | 73.84 | 74.576 |
| Indonesia rupiah | 0.0000677 | 0.0000689 | 14770 | 14522 |
| Japan yen | 0.00926 | 0.00956 | 107.98 | 104.63 |
| 1-month forward | 0.00936 | 0.00948 | 106.88 | 105.49 |
| 3-month forward | 0.00938 | 0.00950 | 106.58 | 105.22 |
| 6-month forward | 0.00941 | 0.00953 | 106.24 | 104.91 |
| Kazakhstan tenge | 0.00246 | 0.0025 | 406.55 | 400.63 |
| Macau pataca | 0.1250438 | 0.1249375 | 7.9970 | 8.0040 |
| Malaysia ringgit | 0.2337 | 0.2341 | 4.2785 | 4.2725 |
| New Zealand dollar | 0.6068 | 0.6087 | 1.6480 | 1.6428 |
| Pakistan rupee | 0.00636 | 0.00628 | 157.15 | 159.25 |
| Philippines peso | 0.0196 | 0.0195 | 51.07 | 51.32 |
| Singapore dollar | 0.707 | 0.7091 | 1.4145 | 1.4102 |
| South Korea won | 0.0008252 | 0.0008256 | 1211.78 | 1211.21 |
| Sri Lanka rupee | 0.005423 | 0.0054684 | 184.4 | 182.87 |
| Taiwan dollar | 0.03317 | 0.03313 | 30.14 | 30.18 |
| Thailand baht | 0.03138 | 0.03152 | 31.87 | 31.73 |
| Vietnam dong | 0.00004308 | 0.00004311 | 23211 | 23197 |
| **Europe** | | | | |
| Bulgaria lev | 0.56567 | 0.57251 | 1.7680 | 1.7470 |
| Croatia kuna | 0.1466 | 0.1471 | 6.8234 | 6.799 |
| Czech Rep. koruna | 0.0422 | 0.04267 | 23.697 | 23.438 |
| Denmark krone | 0.1486 | 0.1497 | 6.7290 | 6.6812 |
| Euro area euro | 1.1104 | 1.1185 | 0.9006 | 0.8941 |
| 1-month forward | 1.1090 | 1.1102 | 0.9017 | 0.9008 |
| 3-month forward | 1.1120 | 1.1130 | 0.8993 | 0.8985 |
| 6-month forward | 1.1150 | 1.1163 | 0.8969 | 0.8958 |
| Hungary forint | 0.00327675 | 0.00330557 | 305.18 | 302.52 |
| Iceland krona | 0.007404 | 0.007492 | 135.07 | 133.48 |
| Norway krone | 0.0994 | 0.0984 | 10.0583 | 10.1657 |
| Poland zloty | 0.2538 | 0.2555 | 3.9403 | 3.9142 |
| Romania leu | 0.2301 | 0.2318 | 4.3456 | 4.3140 |
| Russia ruble | 0.01377 | 0.01335 | 72.623 | 74.905 |
| Sweden krona | 0.103 | 0.1026 | 9.7050 | 9.7493 |
| Switzerland franc | 1.0521 | 1.0594 | 0.9505 | 0.9439 |
| 1-month forward | 1.0502 | 1.0507 | 0.9522 | 0.9518 |
| 3-month forward | 1.0536 | 1.0541 | 0.9491 | 0.9487 |
| 6-month forward | 1.0576 | 1.0581 | 0.9455 | 0.9451 |
| Turkey lira | 0.1579 | 0.1583 | 6.3337 | 6.3168 |
| Ukraine hryvnia | 0.0383 | 0.0387 | 26.1356 | 25.8563 |
| U.K. pound | 1.2284 | 1.2569 | 0.8141 | 0.7956 |
| 1-month forward | 1.2415 | 1.2555 | 0.8055 | 0.7965 |
| 3-month forward | 1.2425 | 1.2564 | 0.8048 | 0.7960 |
| 6-month forward | 1.2428 | 1.2566 | 0.8046 | 0.7958 |
| **Middle East/Africa** | | | | |
| Bahrain dinar | 2.6489 | 2.6524 | 0.3775 | 0.3770 |
| Egypt pound | 0.0637 | 0.0637 | 15.7007 | 15.7011 |
| Israel shekel | 0.2728 | 0.2711 | 3.6662 | 3.6887 |
| Kuwait dinar | 3.2457 | 3.2608 | 0.3081 | 0.3067 |
| Qatar rial | 0.2754 | 0.2762 | 3.6312 | 3.6202 |
| Saudia Arabia riyal | 0.2664 | 0.2664 | 3.7533 | 3.7538 |
| South Africa rand | 0.0614 | 0.0605 | 16.2771 | 16.5405 |

© 2022 McGraw-Hill Education.

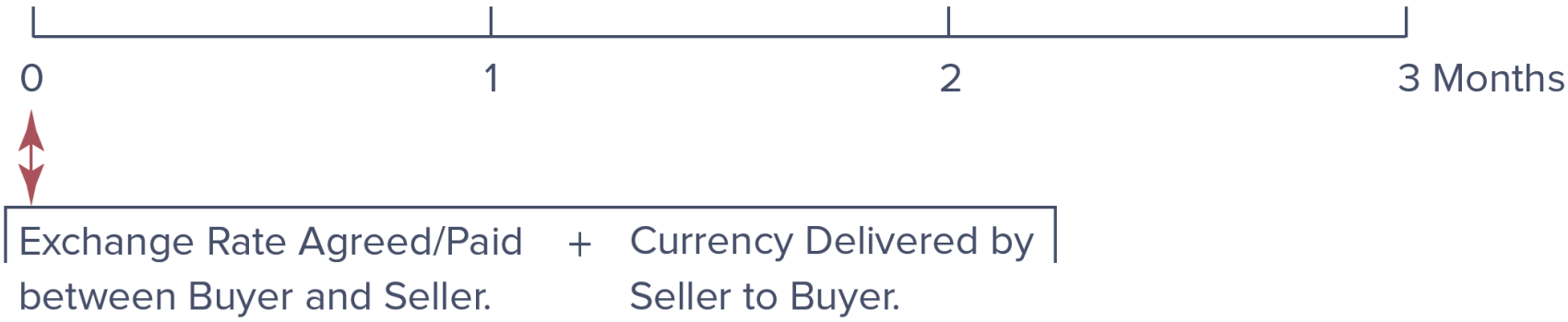
# Foreign Exchange Transactions

- Two types of FX rates and FX transactions:
  1. **Spot FX transactions** involve the immediate exchange of currencies at the current (or spot) exchange rate
  2. **Forward FX transaction** is the exchange of currencies at a specified exchange rate (or forward exchange rate) at some specified date in the future
     - Example is an agreement today (at time 0) to exchange dollars for pounds at a given (forward) exchange rate in three months
     - Typically written for one-, three-, or six-month periods
- Of the $6.6 trillion in average daily trading volume in the FX markets in 2019, 30.1% involved spot transactions while 69.9% involved forward and other transactions

© 2022 McGraw-Hill Education.

# Spot versus Forward Foreign Exchange Transaction

Spot foreign exchange transaction:




Forward foreign exchange transaction:

© 2022 McGraw-Hill Education.

# Exchange Rate of the U.S. Dollar with Various Foreign Currencies

© 2022 McGraw-Hill Education.

# Purchasing Power Parity (PPP)

- **Purchasing power parity (PPP)** is the theory explaining the change in foreign currency exchange rates as inflation rates in the countries change

$$i_{US} = IP_{US} + RFR_{US} \qquad i_S = IP_S + RFR_S$$

where

$i_{US}$ = Interest rate in the United States
$i_S$ = Interest rate in Switzerland (or another foreign country)
$IP_{US}$ = Inflation rate in the United States
$IP_S$ = Inflation rate in Switzerland (or another foreign country)
$RFR_{US}$ = Real risk-free rate in the United States
$RFR_S$ = Real risk-free rate in Switzerland (or another foreign country)

© 2022 McGraw-Hill Education.

# Purchasing Power Parity (PPP) (Continued)

- Assuming real rates of interest (or rates of time preference) are equal across countries:

$$RFR_{US} = RFR_S$$

- Then, the following is true: $i_{US} - i_S = IP_{US} - IP_S$
- PPP theorem states the change in the exchange rate between two countries' currencies is proportional to the difference in their inflation rates:

$$IP_{US} - IP_S = \Delta S_{US/S} / S_{US/S}$$

where

$S_{US/S}$ = Spot exchange rate of U.S. dollars for Swiss francs (or another currency)

© 2022 McGraw-Hill Education.

# Purchasing Power Parity (PPP) (Concluded)

- The **law of one price** is an economic rule which states that, in an efficient market, identical goods and services produced in different countries should have a single price
  - This is the theory behind purchasing power parity
- An example of this line of thinking is demonstrated in *The Economist*'s "Big Mac" index

© 2022 McGraw-Hill Education.

# Interest Rate Parity

- **Interest rate parity theorem (IRPT)** is the theory that the domestic interest rate should equal the foreign interest rate minus the expected appreciation of the domestic currency
- IRPT can be expressed as the following:

$$1 + i_{USt} = (1/S_t) \times (1 + i_{UKt}) \times F_t$$

Return on U.S. investment = Hedged return on foreign (UK) investment

where

$1 + i_{USt}$ = 1 plus the interest rate on a U.S. investment maturing at time $t$

$1 + i_{UKt}$ = 1 plus the interest rate on a UK investment maturing at time $t$

$S_t$ = \$/£ spot exchange rate at time $t$

$F_t$ = \$/£ forward exchange rate at time $t$

© 2022 McGraw-Hill Education.

# Interest Rate Parity: Example

**EXAMPLE 9–6** **An Example of the Interest Rate Parity Theorem**

Suppose that on March 13, 2020, a U.S. citizen has excess funds available to invest in either U.S. or British bank time deposits. It is assumed that both types of deposits are credit- or default-risk free and that the investment horizon is one month. The interest rate available on British pound one-month time deposits, $i_{UK}$, is 0.5 percent monthly. The spot exchange rate of U.S. dollars for British pounds on March 13, 2020 (from Table 9–1), is \$1.2284/£, and the one-month forward rate is \$1.2415/£. According to the IRPT, the interest rate on comparable U.S. one-month time deposits should be:

$$1 + i_{US} = (1/1.2284) \times (1 + 0.005) \times 1.2415 = 1.015718$$

or 1.5305 percent. We can rearrange this relationship as shown above as:

$$\frac{0.015718 - 0.005}{1 + 0.005} = \frac{1.2415 - 1.2284}{1.2284}$$

$$0.010664 = 0.010664$$

Thus, the discounted spread between domestic and foreign interest rates is, in equilibrium, equal to the percentage spread between forward and spot exchange rates.

© 2022 McGraw-Hill Education.